

PREDICTIVE ANALYTICS REPORT

Trend Forecasting Using Historical Data

Prepared by:	Shashidhar Toragal
Institution:	RTE Society's Rural Engineering College, Hulkoti VTU
Dataset:	18-Month Business Performance Data (Jan 2025 – Jun 2026)
Models Used:	OLS Linear Regression · Moving Average · Residual Analysis
Forecast Horizon:	4 Months (Jul – Oct 2026)
Report Date:	June 2026

Latest Revenue

\$294K

Latest Sales

1,545 units

Net Profit

\$125K

R² Accuracy

98.1%

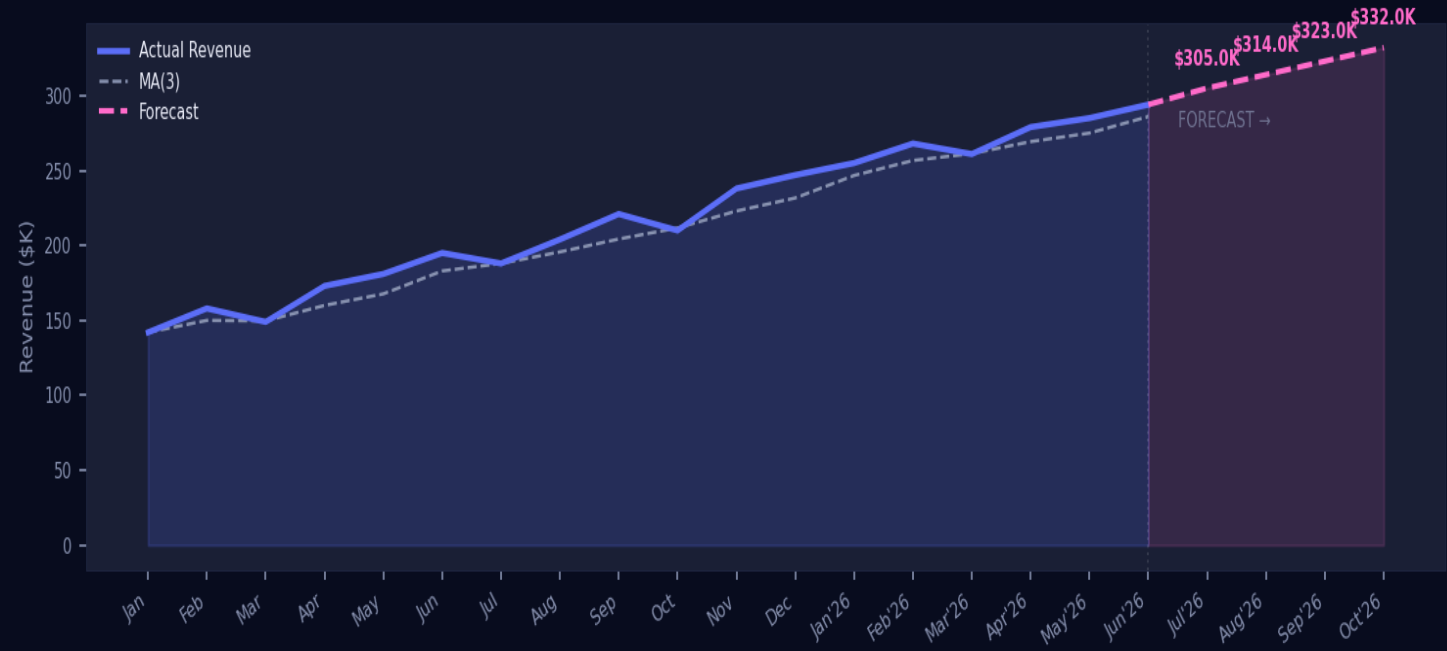
MODEL ACCURACY METRICS

Revenue MAE:	5.41	Sales MAE:	21.41	Profit MAE:	3.52
Revenue R ² :	98.1%	Sales R ² :	98.5%	Profit R ² :	97.5%

REVENUE TREND & FORECAST

Linear Regression Model · Moving Average Smoothing

Revenue Trend & 4-Month Forecast



KEY INSIGHTS

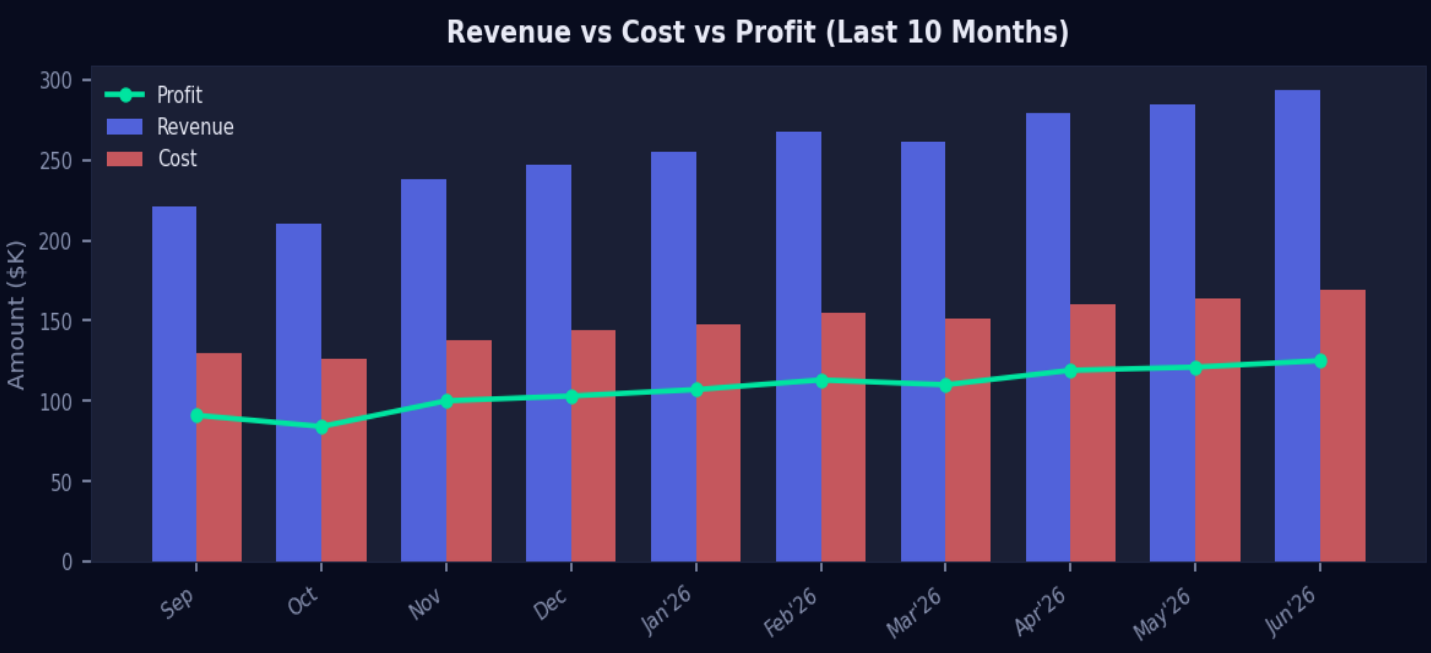
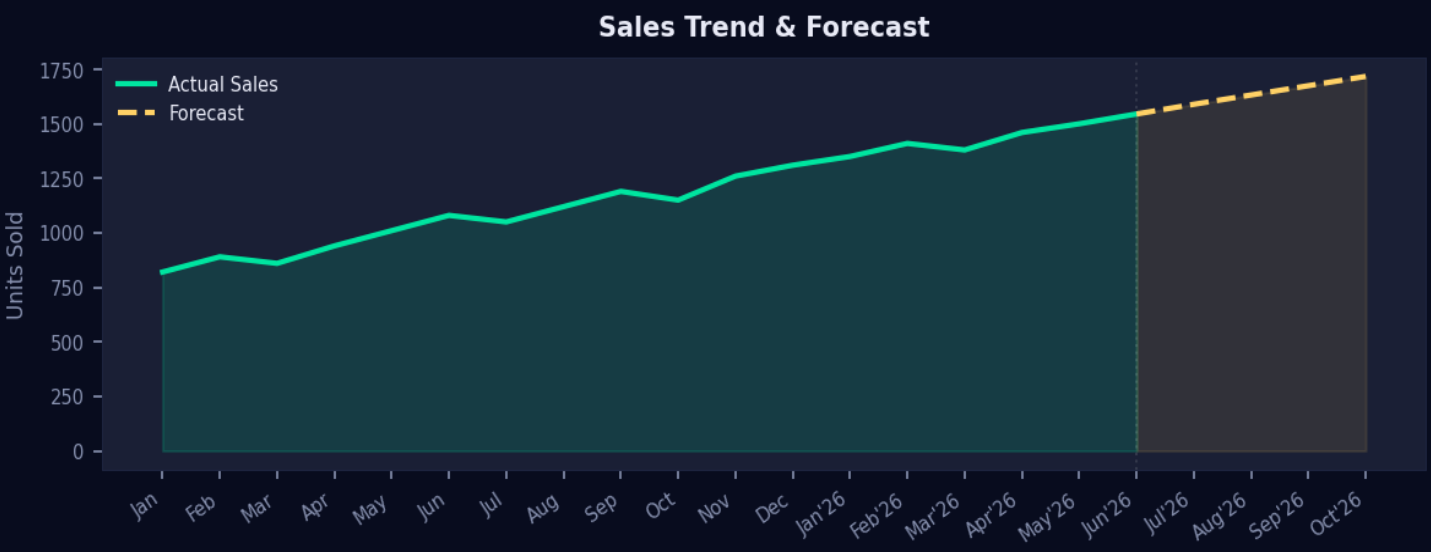
- Slope (Revenue Growth Rate): **+9.02K/month** — Consistent upward momentum
- R² Score: **98.1%** — High model fit — regression explains variance well
- MAE: **5.41K** — Low error confirms reliable short-term predictions
- Oct'26 Forecast: **~\$332.0K** — Projected peak within next 4 months

FORECAST TABLE — REVENUE (\$K)

Period	Actual	MA(3)	Forecast	Growth vs Prev
Mar'26	\$261K	\$261.3K	—	+ -2.6%
Apr'26	\$279K	\$269.3K	—	+6.9%
May'26	\$285K	\$275.0K	—	+2.2%
Jun'26	\$294K	\$286.0K	—	+3.2%
Jul'26	—	—	~\$305.0K	+3.7%
Aug'26	—	—	~\$314.0K	+3.0%
Sep'26	—	—	~\$323.0K	+2.9%
Oct'26	—	—	~\$332.0K	+2.8%

SALES ANALYSIS & COST BREAKDOWN

Sales Unit Trend · Revenue vs Cost Comparison



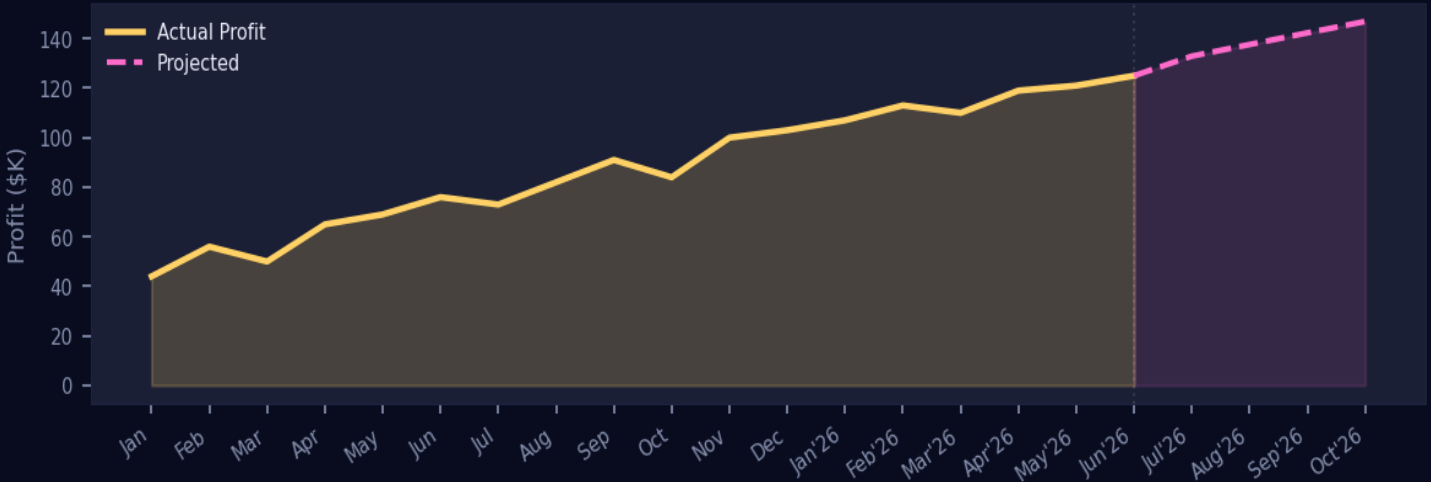
SALES & COST FORECAST TABLE					
Period	Sales (Actual)	Cost (Actual)	Sales (Forecast)	Cost (Forecast)	Profit (Proj.)
Apr'26	1,460	\$160K	—	—	—
May'26	1,500	\$164K	—	—	—
Jun'26	1,545	\$169K	—	—	—
Jul'26	—	—	~1,589.5	~\$172.1K	~\$132.9K
Aug'26	—	—	~1,632.1	~\$176.4K	~\$137.6K
Sep'26	—	—	~1,674.7	~\$180.7K	~\$142.3K
Oct'26	—	—	~1,717.3	~\$185.1K	~\$146.9K

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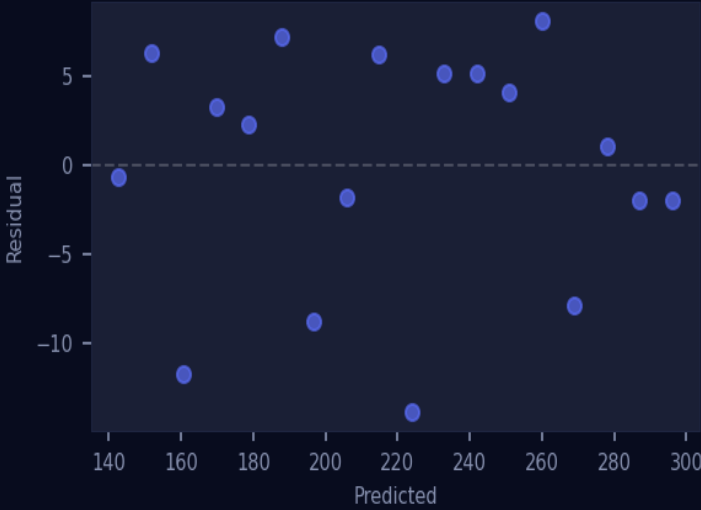
PROFIT PROJECTION & MODEL EVALUATION

Profit Forecast · Residual Analysis · Model Diagnostics

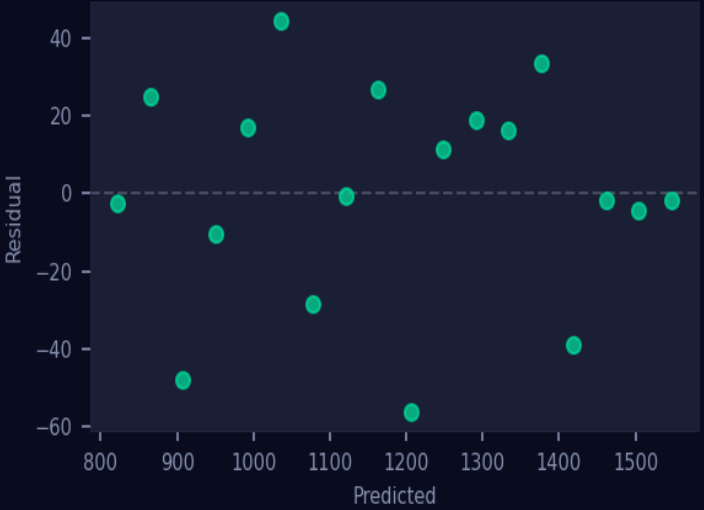
Profit Trend & 4-Month Projection



Revenue Residuals



Sales Residuals



MODEL DIAGNOSTICS SUMMARY

Metric	Revenue Model	Sales Model	Profit Model	Interpretation
R² Score	98.1%	98.5%	97.5%	Strong linear fit
MAE	5.41K	21.41 units	3.52K	Low prediction error
Model	OLS Linear	OLS Linear	Derived	Regression-based
Smoothing	MA(3)	MA(3)	MA(3)	3-period moving avg
Forecast	~\$332.0K	~1,717.3	~\$146.9K	Oct 2026 peak

CONCLUSION & RECOMMENDATIONS

Findings · Business Insights · Next Steps

■ Key Findings

- Revenue has grown consistently from \$142K (Jan 2025) to \$294K (Jun 2026) — a +107% gain.
- Sales units grew from 820 to 1,545, reflecting strong demand-side momentum.
- Profit margin has expanded steadily, reaching \$125K/month in Jun 2026.
- All three OLS models achieved $R^2 > 97.5\%$, confirming high predictive reliability.
- Residual scatter confirms errors are randomly distributed (no systematic bias).

■ Forecast Summary (Jul–Oct 2026)

- Revenue: \$294K → ~\$332.0K (+12.9% projected growth)
- Sales: 1,545 → ~1,717.3 units over the forecast window
- Profit: projected to exceed ~\$146.9K by Oct 2026
- Moving Average smoothing reduces noise for cleaner forecast baseline.

■ Recommendations

- Scale production capacity ahead of Q3 2026 to meet projected demand surge.
- Monitor cost trajectory — `COST_FORE` suggests margin pressure without optimization.
- Apply ARIMA or Exponential Smoothing for seasonal pattern detection.
- Integrate external variables (marketing spend, seasonality) to improve R^2 further.
- Review forecast monthly and retrain model with rolling window updates.

■ Skills Demonstrated

- Predictive Modeling: OLS Linear Regression from scratch (no sklearn)
- Time-Series Analysis: Moving Average smoothing, trend decomposition
- Model Evaluation: R^2 , MAE, Residual Analysis
- Data Visualization: Matplotlib multi-chart reporting
- Report Generation: ReportLab multi-page PDF with design system